

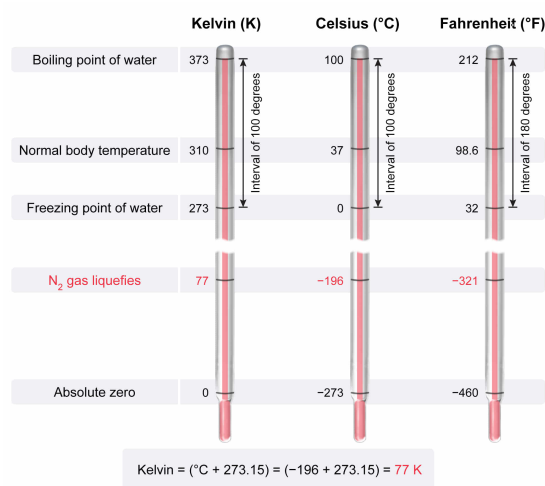
Liquid nitrogen is used in cryosurgery for wart removal. The liquefaction of N_2 gas begins to occur at $-196\text{ }^\circ\text{C}$ and liquid N_2 freezes at $-210\text{ }^\circ\text{C}$. Which temperature range coincides with the onset of the condensation of N_2 gas?

- ☐ A. 61–65 K [12%]
✔ ☒ B. 75–79 K [76%]
☐ C. 467–471 K [9%]
☐ D. 481–485 K [1%]

Correct

75% answered correctly

Explanation:



The **Kelvin scale** differs from the **Celsius scale** only in the placement of the zero value. The Celsius scale defines the zero value with respect to the freezing point of water whereas the Kelvin scale moves the zero value to be defined with respect to absolute zero ($-273.15\text{ }^\circ\text{C}$), which is the coldest temperature theoretically possible. Accordingly, $0\text{ K} = -273.15\text{ }^\circ\text{C}$, and the Kelvin scale is considered an absolute temperature scale because it has no negative values.

Because the size of the units are the same on both the Celsius and the Kelvin scales and the scales differ only by the placement of the zero value, a **change of 1 degree Celsius is equal to a change of 1 Kelvin**. As a result, a temperature given in degrees Celsius can be converted to its equivalent on the Kelvin scale by adding 273.15 (often rounded to just 273). Similarly, the conversion of Kelvin temperatures to degrees Celsius is done by subtracting 273.15 from the Kelvin temperature.

Nitrogen gas (N_2) begins to undergo liquefaction (**condensation**) near $-196\text{ }^\circ\text{C}$. On the Kelvin scale, this value is found to be

$$\text{Kelvin} = (^\circ\text{C} + 273) = (-196\text{ }^\circ\text{C} + 273) = 77\text{ K}$$

Therefore, a temperature range of 75–79 K is an equivalent temperature range on the Kelvin scale that coincides with the onset of the condensation of N_2 gas.

(Choice A) Liquid N_2 freezes at $-210\text{ }^\circ\text{C}$ (63 K). A temperature range of 61–65 K coincides with the freezing point of liquid N_2 rather than the condensation of N_2 gas.

(Choices C and D) Conversion of temperatures in Celsius to the Kelvin scale requires *adding* 273 to the Celsius values. If the temperatures $-196\text{ }^\circ\text{C}$ and $-210\text{ }^\circ\text{C}$ are instead subtracted from 273, erroneous values of 469 K and 483 K, respectively, are obtained.

Educational objective:

The Kelvin and Celsius scales differ only in the placement of the zero value. Therefore, a change of 1 degree Celsius is equal to the change of 1 Kelvin. A temperature on the Celsius scale can be converted to its equivalent on the absolute Kelvin scale by adding 273.15 to the Celsius temperature.

Time spent:0 sec

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System:
Thermodynamics, Kinetics & Gas Laws